ChatGPT's Information Seeking Strategy: Insights from the 20-Questions Game

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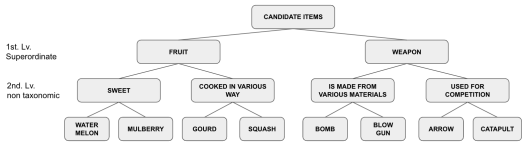
Abstract

Large Language Models, and ChatGPT in particular, have recently grabbed the attention of the community and the media. Having reached high language proficiency, attention has been shifting toward its reasoning capabilities. In this paper, our main aim is to evaluate ChatGPT's question generation in a task where language production should be driven by an implicit reasoning process. To this end, we employ the 20-Questions game, traditionally used within the Cognitive Science community to inspect the information seeking-strategy's development. This task requires a series of interconnected skills: asking informative questions, stepwise updating the hypothesis space, and stopping asking questions when enough information has been collected. We build hierarchical hypothesis spaces, exploiting feature norms collected from humans vs. ChatGPT itself, and we inspect the efficiency and informativeness of ChatGPT's strategy. Our results show that ChatGPT's performance gets closer to an optimal agent only when prompted to explicitly list the updated space stepwise.



Questioner	Answerer	Sp
1. Is it a weapon?	No	4
2. Is it sweet?	Yes	2
3. Is it the watermelon?	No	1
4. Is it the mulberry?	Yes	

Figure 1: **Upper part**: an example of a Hierarchical Hypothesis Space built with ChatGPT-feature norms. **Bottom part**: an example of an optimal Questioner which always divide the space (Sp) into half (starting with 8 candidates, going to 4:4, and then 2:2). Questions at turns 1-2 are constraint-seeking (CS), while the 3rd and 4th are hypothesis-scanning (HS). With the halfsplit procedure, the target can be guessed with just 3 turns or at most with 4, as in the example.

1 Introduction

ChatGPT's impressive ability to solve numerous natural language tasks has put it in the spotlight of Academia and media attention (Bang et al., 2023; Laskar et al., 2023). The success on a variety of tasks has brought people to even claim that GPT-4 "could reasonably be viewed as an early (yet still incomplete) version of an artificial general intelligence (AGI)" (Bubeck et al., 2023). Others are more cautious, showing the weakness of the model's reasoning abilities, (e.g., Bang et al. 2023).

A core aspect of human intelligence is the implicit connection between the reasoning process and language production. This connection strongly drives the generation of questions in information-seeking scenarios which, therefore, have been largely studied in Cognitive Science. After the pioneering work by Mosher and Hornsby (1966), the 20-Questions game has been employed to observe children's cognitive developmental trajectory: A player thinks of an entity, the second player is given a set of candidates (e.g., *cat*, *dog*, *bird*) and has to identify the target entity among the possible candidates by making Yes-or-No questions. This and following experiments have shown that through the developmental trajectory, children learn to recognize object-general features, cluster similar objects into categories and use such categorization to ask context dependent informative questions: they shift from *Hypothesis Scanning* questions ("*Is it a dog?*") to *Constraint-Seeking* questions ("*Does it has four legs?*"). Such a shift let the elder children be more efficient in their information seeking process. Moreover, pre-scholar children tend to continue asking questions when enough information has been collected (i.e., the space has reduced to one candidate). They do not know when to stop (Ruggeri et al., 2016), a core skill of infor-

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mation search and decision-making (Todd et al., 1999). Interestingly, Ruggeri et al. (2021) uses a hierarchical version of the 20-Q game, in which candidates are organized into three category levels based on shared features; by providing children with the object-related features needed to halfs-plit the space, children were able to target such higher category levels, reaching the solution more efficiently. Inspired by this literature, we use a hierarchical 20-Q game to evaluate whether ChatGPT is able to generate questions driven by its reasoning over the Hypothesis Space (HypSp).

We leverage on the widely used feature norms elicitated from human annotators, McRae-norms (McRae et al., 2005) to build the hierarchical hypothesis spaces. Such norms reflect humans' knowledge representation which could differ from ChatGPT's knowledge. To mitigate this potential difference, we build also hypothesis spaces using norms elicitated from ChatGPT itself, (GPT-norms (Hansen and Hebart, 2022)). Figure 1 reports an example of an 8 candidate symmetric nested space based on GPT-norms.

We prompt ChatGPT[1] to play the 20-Q game, both in the role of the Questioner and of the Answerer.[2] We aim to understand whether the (a) Questioner is able to identify the high-level property that clusters the space and hence asks whether the target has that property; (b) it also knows whether all the other candidates have or do not have that property, and is able to use such information to update the HypSp stepwise; and finally (c) it understands when to stop asking questions, i.e., the HypSp is reduced to a singleton. Figure 1 includes a dialogue an optimal agent could ask, if driven by an half-split search. Our results show that ChatGPT's performance is far from an optimal agent when having to update the space internally and it is closer to it when prompted to explicitly list the updated space stepwise.

2 Related Work

Our work put together two research lines: the current effort of the AI community to evaluate ChatGPT language and reasoning skills, and the cognitive science literature focusing on the developmental trajectory of information search strategies in humans.

[1]We used the API version of `gpt-3.5-turbo` available between March and May.

[2]The data and scripts associated to this paper are available at `https://github.com/leobertolazzi/20q-chatgpt`.

ChatGPT evaluation Bang et al. (2023) run a deep and broad evaluation of ChatGPT on a variety of well recognized benchmarks in the Natural Language Processing community. ChatGPT results to be State-of-the-Art in zero-shot setting for most natural language understanding tasks. Though it is more suitable for open-domain dialogue tasks, it performs well also in task-oriented dialogues, and it is able to keep track of information given in previous turns, when answering follow up questions. Moreover, Bang et al. (2023) evaluate ChatGPT reasoning skills: though it lacks inducting reasoning skills, it performs well on deductive reasoning in clean settings. However, as other LLMs (Ott et al., 2023), ChatGPT as well encounters problems with complex deductive reasoning involving multi-hops, viz. a combination of facts spread in different passages of a corpus. Zhu et al. (2023) challenged ChatGPT on the Visual Dialogue task, originally proposed by Das et al. (2017). The informativeness of the question is measured on the quality of the caption it summarises out of the dialogue. As far as we know, this is the first work to evaluate the information seeking strategy of a LLM using the 20 Questions game. Our research question is whether and to what extent the language generation of a LLM is tied to reasoning.

Developmental and Cognitive Psychology Starting with Mosher and Hornsby (1966), the 20-Q search task has been largely used in developmental and cognitive psychology. Among the measures to evaluate the question's informativeness, Expected Information Gain (Lindley, 1956) emerges as one of the most used. It values questions with respect to the uncertainty reduction, and it is usually connected with the prior probability. Subjects have been evaluated with the 20-Q game considering both scenarios simulating *prior* expectations and scenarios with uniform distribution (e.g., Ruggeri and Lombrozo 2015; Meder et al. 2019; Ruggeri et al. 2021; Testoni 2023). Our scenario is the uniform distribution.

It is widely accepted that children's search strategies are less efficient than adults' ones. Rather than identifying high-order properties splitting efficiently the Hypothesis Space, indeed, children tend to scan the space item by item. In complex scenarios, it has been shown that adults do not efficiently plan ahead; they tend to follow a half-splitting strategy: ask the question that more closely approximates a division of the space into half (Meder

et al., 2019). Rothe et al. (2018) show that people can accurately evaluate questions quality, but have limited ability to optimize the informativeness of their questions. By leverage of feature norm collections, we work on a simplified scenario where adults would more easily stay close to an optimal agent.

Feature Norms Feature norms refer to minimal semantic descriptions that capture the typical attributes associated with a collection of objects or concepts (e.g., a *dog* can be described by features such as *has fur* and *does bark*). One common method of acquiring semantic features for concepts is to ask individuals to list properties associated with a given concept. A broadly used collection is the McRae-norms (McRae et al., 2005) which comprise 2524 unique features collected from approximately 725 participants, which are in turn categorized according to Wu and Barsalou (2009)'s taxonomy of relations (WB). These norms encompass 541 animate and inanimate concrete concepts, with an average of 30 participants providing feature listings for each of them. The features included in the McRae-norms are of various types, such as physical (perceptual) properties, functional properties, taxonomic properties, and encyclopedic facts. Inspired by this work and to obtain a large-scale collection, (Hansen and Hebart, 2022) instructed GPT-3 (Brown et al., 2020) to generate semantic feature norms for a diverse set of 1,854 concrete concepts which have been annotated with 84561 unique features elicitated from the model through 30 runs, pre-processed and filtered. These feature norms were then released by the authors for public use and exploration; the authors expanded their method to other models of the GPT family when they became available. We use the feature norms obtained from GPT-3.5-turbo, and refer to them as GPT-norms.[3]

3 Hierarchical version of the 20-Q game

Following Ruggeri et al. (2021), we created a hierarchical version of the 20-Q game. In other words, the hypothesis spaces are built out of two subsets of equal size (N:N), and iteratively divided into further subsets based on some other features. We exploit McRae (McRae et al., 2005) and ChatGPT (Hansen and Hebart, 2022) feature norms to build the nested structures. We consider hierarchies of two levels (8 candidates, divided into 4:4, and 2:2) and of three levels (16 candidates, divided into 8:8, 4:4, and 2:2). The first level is always based on superordinate properties ($F_{1_a} vs.\ F_{1b}$), which are by definition mutually exclusive (e.g., *bird* vs. *mammal*, *fruit* vs. *weapon*, etc.). The subsets of the other levels instead are obtained from all the other feature norms associated with the candidates (e.g., items that are *fruit* could be divided into those that are *sweet* vs. those that are *cooked in various way*). We make sure that the feature that is shared by half of the candidates is not listed for any of the item in the other half, and viceversa. The leaves of the hierarchy are randomly selected among the concepts of the corresponding groups. See Figure 1 for the schema and an example with 8 candidates organized based on GPT-norms.

Hierarchical Hypothesis Space creation Our starting point are the concepts in McRae et al. (2005) and in Hansen and Hebart (2022), 541 and 1854, respectively. For McRae-norms, we selected the superordinates frequent enough to let us create spaces of 8 and of 16 candidates.[4] For the second level, we use features of the other 8 most frequent WB relations (51 unique features). For the Hypothesis Spaces built from ChatGPT-norms, we selected the same 6 superordinates for the first level splits, and other 806 most frequent unique features for the second levels. We built the hypothesis space through a recursive process that guarantees variety and randomness of the selection (See the Supplementary Material for details). We will refer to these two types of hypothesis spaces as 8 vs. 16 candidates sets (cds), distinguishing the former into McRae- and GPT-based 8 cds; henceforth, `8-McRae`, `16-McRae`, `8-GPT`.

Game creation A game consists of a set of candidates, assigned to the Questioner player, and a target among them, assigned to the Answerer. We build 90 games for each of the three types of Hypothesis Space as follows. First of all, out of the 6 selected superordinate features, we build all possible pairs, viz. 15 (F_{1a}, F_{1b}); we then randomly select 6 sets of candidates for each of the 15 pairs, yielding 90 unique sets (total 270 sets). Finally, we build the 90 games by randomly selecting the target 3 times from the candidates that share F_{1a}

[3]The norms collected with `gpt-3.5-turbo` are available at `https://github.com/ViCCo-Group/semantic_features_gpt_3`.

[4]The 6 superordinates we use to build the first level splits are: mammal, bird, clothing, weapon, fruit, vegetable.